Yue Ma

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PROFILE

Research Engineer and Physics Ph.D. building cost-aware agentic systems that automate long-horizon experimental loops in machine learning and science. Develops evaluation frameworks and trajectory infrastructure to capture, analyze, and improve scientific decision-making under real compute constraints.

EDUCATION & ACADEMIC POSITIONS

Postdoctoral Researcher, Halıcıoğlu Data Science Institute, UC San Diego — Mar 2026 — Present

Ph.D. in Particle Physics, University of California, San Diego — Oct 2020 — Feb 2026

B.S. in Physics, Nanjing University — Aug 2016 — Jun 2020

AGENTIC SYSTEMS, EVALUATION, AND RESEARCH INFRASTRUCTURE

SIDERIUS: Resource-Aware Multi-Agent System for Autonomous Research — Feb 2026 — Present

A closed-loop agentic system for proposing, implementing, and evaluating machine-learning experiments under online compute constraints. Sole architect and developer; initially deployed on scientific time-series denoising and now extending to additional domains.

— **Multi-Agent Research Runtime:** Designed a typed 6-agent workflow spanning literature review, hypothesis generation, PyTorch implementation, validation, experimentation, and interpretation, with structured findings and failure evidence propagated across iterations.

— **Online Resource Control:** Built dynamic runtime and GPU-memory calibration, isolated per-candidate measurement, and admission control across trial-, formal-, and campaign-level compute budgets.

— **Reliable Long-Horizon Execution:** Implemented scientific-quality gates, watchdogs, bounded retries, fail-closed budget enforcement, and resumable state; achieved over a 5× denoising improvement during autonomous campaigns on the TIDMAD benchmark.

SciTra: Cost-Aware Scientific-Agent Evaluation and Trajectory Infrastructure — Jul 2026 — Present

Founder and lead of a 50+ scientist, 20+ institution collaboration developing scientific-agent evaluations and decision-trajectory infrastructure.

— **Cost-Aware Scientific Evaluation:** Formalized open-ended tasks around expensive iterative decisions, treating experiment cost rather than idea generation as the primary scaling bottleneck; coordinated 30+ task ideas across 11 domains.

— **Scientific Decision Trajectories:** Built a cross-vendor runtime that captures expert—agent interactions, tool use, Git changes, failures, and explicit cost, output, and decision records as typed trajectories of how scientific choices are made and revised.

— **Auditable Research Infrastructure:** Implemented append-only declarations, frozen evidence, integrity manifests, interrupted-run recovery, and deterministic export for reproducible trajectory evaluation.

FrontierPhysics: Benchmark for Iterative Frontier Physics Research (Core Team) — Jul 2026 — Present

Core team member on a benchmark, built with the BenchFlow team, that evaluates how AI agents carry out frontier physics research iteratively.

— **Iterative Research Tasks:** Tasks spanning literature review through research-plan implementation, drawn from problems that take PhD-level researchers weeks rather than from questions with a single retrievable answer.

— **Cross-Institution Task Development:** Built with contributors across universities, national laboratories, and industry, so that task difficulty is set by working physicists rather than by benchmark authors.

MACHINE LEARNING RESEARCH

Adaptive Neural Processes for Extreme Forecasting — Nov 2024 — Jun 2025

Probabilistic forecasting under distribution shifts and limited observations. arXiv:2602.04609

- **Model Development:** Conceived and derived AdaCNP, a target-conditioned context-reweighting mechanism that enables few-shot adaptation to unseen extreme regimes without retraining.
- **Experimental Evaluation:** Achieved a 22% reduction in MSE and the best negative log-likelihood among Neural Process baselines across two real-world power-system datasets.

Multi-Fidelity Surrogate Modeling for Scientific Simulation — Jul 2024 — Oct 2024

Learning efficient surrogates for computationally expensive, high-variance detector simulations.

- **Scientific ML System:** Developed a multi-fidelity surrogate combining Gaussian Processes and Neural Processes to emulate high-dimensional rare-event simulations.
- **Compute-Efficient Optimization:** Reduced simulation cost by 96.7% while improving background-suppression efficiency by 66.5%; contributed core algorithms and derivations to the ICLR 2025 Spotlight paper "RESuM" (acknowledged for significant contribution).

STATISTICAL INFERENCE AND LARGE-SCALE SCIENTIFIC COMPUTING

Research within the XENONnT Collaboration, a 200+ scientist international experiment operating petabyte-scale data systems for dark-matter and solar-neutrino searches.

Probabilistic Modeling and Statistical Inference (Analysis Lead) — Oct 2020 — Present

Leading a team of ~20 scientists developing the first measurement of solar pp neutrinos at ultra-low momentum transfer.

- **Inference and Calibration:** Developed Bayesian and maximum-likelihood inference frameworks, improving the precision of efficiency estimates by 60% through new calibration strategies and data-selection methods.
- **Experimental Evaluation:** Built Monte Carlo workflows for uncertainty propagation, sensitivity estimation, goodness-of-fit testing, and hypothesis testing, informing analysis design and experimental strategy.

Data Infrastructure and High-Throughput Computing — Jan 2024 — Present

- **Distributed Data Systems:** Key maintainer of the open-source strax and straxen frameworks processing petabyte-scale detector data for 200+ collaborators.
- **Performance Optimization:** Reduced ETL processing latency by 40% through vectorized NumPy and Pandas workflows, Numba acceleration, and memory-efficient data processing.

TECHNICAL SKILLS

Programming & ML	Python, PyTorch, NumPy, Pandas, SciPy, Scikit-learn, C/C++, Bash, Linux, Git
Agents & Evaluation	LLM agent orchestration, autonomous experimental loops, resource-aware evaluation, trajectory instrumentation, failure diagnostics, multi-agent systems
Systems & Infrastructure	Pydantic, FastAPI, pytest, GitHub Actions, SLURM/HPC, ETL pipelines, resource monitoring and performance optimization

SELECTED PUBLICATIONS

- Chenxi Hu, **Yue Ma**, Yifan Wu, Yunhe Hou (2026). "Resilient Load Forecasting under Climate Change: Adaptive Conditional Neural Processes for Few-Shot Extreme Load Forecasting". arXiv:2602.04609.
- XENON Collaboration, E. Aprile *et al.* (2023). "First Dark Matter Search with Nuclear Recoils from the XENONnT Experiment". *Physical Review Letters* **131**, 041003.
- XENON Collaboration, E. Aprile *et al.* (2022). "Search for New Physics in Electronic Recoil Data from XENONnT". *Physical Review Letters* **129**, 161805.